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A Review of the Validity and Reliability of Accelerometer-Based Metrics From Upper Back–Mounted GNSS Player Tracking Systems for Athlete Training Load Monitoring

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Abstract

Dawson, L, Beato, M, Devereux, G, and McErlain-Naylor, SA. A review of the validity and reliability of accelerometer-based metrics from upper back–mounted GNSS player tracking systems for athlete training load monitoring. *J Strength Cond Res* 38(8): e460–e475, 2024—Athlete load monitoring using upper back–mounted global navigation satellite system (GNSS) player tracking is common within many team sports. However, accelerometer-based load monitoring may provide information that cannot be achieved with GNSS alone. This review focuses on the accelerometer-based metrics quantifying the accumulation of accelerations as an estimation of athlete training load, appraising the validity and reliability of accelerometer use in upper back–mounted GNSS player tracking systems, the accelerometer-based metrics, and their potential for application within athlete monitoring. Reliability of GNSS-housed accelerometers and accelerometer-based metrics are dependent on the equipment model, signal processing methods, and the activity being monitored. Furthermore, GNSS unit placement on the upper back may be suboptimal for accelerometer-based estimation of mechanical load. Because there are currently no feasible gold standard comparisons for field-based whole-body biomechanical load, the validity of accelerometer-based load metrics has largely been considered in relation to other measures of training load and exercise intensity. In terms of convergent validity, accelerometer-based metrics (e.g., PlayerLoad, Dynamic Stress Load, Body Load) have correlated, albeit with varying magnitudes and certainty, with measures of internal physiological load, exercise intensity, total distance, collisions and impacts, fatigue, and injury risk and incidence. Currently, comparisons of these metrics should not be made between athletes because of mass or technique differences or between manufacturers because of processing variations. Notable areas for further study include the associations between accelerometer-based metrics and other parts of biomechanical load-adaptation pathways of interest, such as internal biomechanical loads or methods of manipulating these metrics through effective training design.

Key Words: PlayerLoad, Dynamic Stress Load, Body Load, acceleration, biomechanical load

Introduction

Athletic training can have various intended and unintended effects on the human system. The investigation, prescription, and monitoring of these cause-and-effect relationships require the training and the outcome to be quantified. "Training load" is considered a multidimensional construct consisting of 2 causally related subdimensions: "external training load" and "internal training load" (98–100), each of which can be subdivided into physiological or biomechanical load. For example, external physiological load could include kinetic energy (i.e., distance covered), whereas internal physiological load could be oxygen uptake or cardiovascular demand (189). External biomechanical load may include ground reaction forces (GRF) or center of mass (CoM) accelerations, whereas internal biomechanical load may refer to joint contact forces or muscle-tendon forces (189). Identification of an appropriate measure or metric of training load ideally requires a framework of causal relationships between external load measures, internal load measures, and training outcomes (100,114,144,201). These frameworks can be informed by research in physiology, biomechanics, and other areas (7,100,107,108). Determining appropriate measures of training load can provide application in the prescription and monitoring of training sessions within an overall framework of physical training (72,74,102) and inform subsequent session design based on differences between prescribed and achieved training loads (44).

The past 2 decades have seen rapidly increasing use of wearable technologies to monitor athlete training loads (33,88). However, there is a need for practitioners to first refine their requirements for a training load monitoring system within any causal framework and then evaluate available devices against those requirements (8,37). One of the most well-known and commonly used areas of technology for this purpose is global positioning system (GPS) or global navigation satellite system (GNSS) technology (8,20–22,49). Global positioning system is 1 subsection of GNSS, with modern GNSS accessing a greater number of satellites and therefore potentially providing greater precision and reliability (101,123). Both GPS and GNSS are considered collectively in this article as GNSS.

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A GNSS player tracking unit can accurately report total and high-speed distance covered, and peak and average speeds (21) but underestimates low-velocity high-impact events with minimal horizontal displacement, such as tackling, jumping, landing, blocking, and other collisions (5,28). Frequent changes of direction can also affect the validity and reliability of some GNSS metrics, such as total distance (140,155) and instantaneous velocity (94). Applications of GNSS are further limited in indoor sports, although local positioning systems can be used (147,171) and when activity is performed with relatively little translational movement (192). Consequently, an inertial measurement unit (IMU) comprising accelerometer, gyroscope, and magnetometer is often incorporated into the GNSS unit (9,58). The accelerometer provides the capability to monitor external biomechanical load through instantaneous accelerations experienced by the athlete (123,193). As the trunk is the heaviest body segment and positioned proximally on the body, these accelerations have been proposed to represent whole-body CoM accelerations and therefore relate to the GRF or impact magnitudes experienced. However, it may not be possible to accurately predict whole-body CoM accelerations and hence GRF from accelerations of a single or small number of segments (61,137,138,190,203). Such external forces describe the biomechanical loading experienced by the musculoskeletal system and have been commonly investigated in association with running performance (26,136,150) and injuries (188,209). However, relationships between GRF and musculoskeletal injuries are often unclear (188,209), and these external forces may not reflect internal biomechanical loads (126,210).

From a training perspective, the monitoring of instantaneous accelerations is particularly useful for assessing external load in collision sports, such as rugby and some codes of football, given the frequency of impacts (93,96,106). Accelerometer-based metrics include peak acceleration, number of accelerations above a specified threshold, resultant vector quantities, and cumulative acceleration metrics that estimate mechanical load (e.g., PlayerLoad, Dynamic Stress Load, Body Load) (30,153,189). Although coaches, sport scientists, and medical teams are using accelerometer-based metrics to monitor components of performance, training load, fatigue, and injury risk (3,43,49,120,200), it is less apparent which metrics have received adequate assessment of validity and reliability (30,97,121,139). To optimize training for a specific outcome within a causal framework, any applied measure of training load should reflect the mediating mechanisms of the primary outcome (45,100,114). Therefore, the purpose of validating training load metrics is to assess the extent to which this is true for any given metric (100). If a training load measure cannot be connected causally to a plausible mediator of the desired outcome, then it is likely of little practical use for purposes of training optimization (100). Because of the numerous performance or injury outcomes of interest, each with their own set of mediators, there cannot be a single best metric that reflects all causal pathways from athletic training to its outcomes. Ultimately, the appropriateness of a metric may depend on what aspects of a causal framework it is intended to describe or predict, its relationships with other aspects of the causal framework, the acceptability of its limitations, and whether the assumptions on which it is based are reasonable or testable (100). Given that researchers and practitioners are using these metrics, it is important to know what they can and cannot do in a valid and reliable manner as part of a causal framework.

Therefore, the aim of this review is to appraise the literature regarding the validity and reliability of accelerometer use in upper back–mounted GNSS player tracking systems, with a focus on accelerometer-based metrics estimating external biomechanical load from the accumulation of instantaneous accelerations. Specifically, this review aims in assessing the validity, reliability, and potential applications in monitoring athlete load, training prescription, and injury management as a general concept, through accumulating acceleration measurements from GNSS-housed accelerometers. This process involves applying manufacturer-specific algorithms to the accelerometer signal, reported as arbitrary units. Examples include Player Load (Catapult, Melbourne, Australia), Dynamic Stress Load (STATSports, Newry, Northren Ireland) and Body Load (GPSports, Canberra, Australia).

Accelerometers

Accelerometers measure linear acceleration and are typically uniaxial or triaxial, measuring accelerations in 1 or 3 (e.g., anterior-posterior, mediolateral, and vertical) axes, respectively (203). Triaxial microelectromechanical system accelerometers are the most common in sport, measuring acceleration magnitudes in each axis (9,82). These devices can collect data at high frequencies, are sensitive to rapid changes in acceleration (185), and can be used in indoor venues because of the lack of a requirement for satellite connection.

Accelerometers are typically used to measure accelerations of a body segment (e.g., torso, shank) or an object (e.g., racket, stick) (40). Investigations into optimal accelerometer placement have identified CoM (70,110,115,122) or ankle attachments (53,90,125) to be most effective for estimating GRF and thus the external biomechanical load experienced by the athlete (70,80,92,122). This is perhaps unsurprising due to the fundamental relationship between GRF and CoM acceleration (69,111). However, accelerometers embedded in GNSS units are mounted on the upper back for optimal GNSS signal acquisition (18,123) and athlete comfort. Practitioners should consider that upper back–mounted accelerometer signals can be affected by factors, such as incidental harness or attachment movement (61,132,206); therefore, sensor fixation should be as tight as is physically comfortable to minimize additional displacement (32,95,118,186). In addition, trunk positioning (24,68) and gait cycle features, including cadence, step length, and gait regularity and symmetry (134,159), can affect the magnitude and timing of the postimpact elastic wave (propagation of impact-related energy [i.e., measured accelerations] through body tissues and decreasing of amplitude with distance from the source). Finally, muscle damping by which sites further from impact are also less affected by the amplifying effect of greater GRF (35,129,194) Therefore, the correlations between accelerations or accelerometer-based metrics at different body positions may not be particularly high (80). It should also be considered that acceleration signals comprise various frequencies, not all of which relate directly to the postimpact elastic wave (117,129,174).

Accelerometer-Based Load Metrics

Sport and exercise science researchers and practitioners have typically sought to convert continuous signals (e.g., an acceleration time-series signal) into discrete metrics such as mean or peak values, and durations (100). This often sees intensity and duration combined to operationalize a single cumulative training load metric (39,100). However, practitioners often prefer automated metrics from which meaning can be relatively easily inferred, rather than continuous data from an entire session that requires

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complex additional processing and lacks direct application. Metrics would ideally represent an underlying (e.g., mechanical) phenomena and relate to causal mechanisms of injury risk, tissue remodeling, or training adaptations that can be tested as part of a causal pathway through transparent calculation and processing methods. Once an appropriate measure of training load is identified, it is necessary to determine the relevant dimensions (i.e., metrics) for its quantification and application (45,100). Table 1 summarizes what is known of the equation and calculation process for 3 of the most common accelerometer-based metrics: PlayerLoad, Body Load, and Dynamic Stress Load.

Since the introduction of PlayerLoad, the definition and equation used have become less clear. One study found Catapult's automatically derived PlayerLoad values to be 15.1 ± 1.2% lower than those calculated manually using the stated equation (139). Bredt et al. (30) identified 4 differing definitions and 4 differing equations for PlayerLoad between publications (2,27,34,36,154,165,198). Hollville et al. (91) claimed to have determined the manufacturer processing methods during a pilot analysis as low-pass filtering with a zero-phase-shifting third-order Butterworth filter at 1 Hz, applying a scaling factor of 4. Despite these inconsistencies, PlayerLoad has been the most commonly researched metric (81,189).

Variation also exists in the equations reported for GPSports' Body Load metric, as demonstrated in Table 1. As a result, it is not always clear which version of the metric has been used within the literature. GPSports are now under the Catapult umbrella, and Body Load is less frequently used, but devices and systems remain in circulation. Research using Body Load can still inform our understanding of accelerometer-based metrics and their capabilities/limitations within causal frameworks of interest.

The 3 metrics differ in their apparent underlying assumptions or their operationalization of the training load concept. Body Load includes individual acceleration values, similar in some ways to the calculation of GRF impulse; Dynamic Stress Load includes peak accelerations, similar in some ways to the calculation of peak GRF; and PlayerLoad includes the change in acceleration, similar in some ways to the calculation of GRF loading rate. In reality, each metric is more complex than that with multiple components, scaling factors, and nonlinear components. Nonetheless, it remains important to consider how the underlying equations may reflect any hypothetical relationships between "load" and subsequent outcomes of interest. As recently stated by Impellizzeri et al. (100), it appears that the "burden of proof" has been reversed; non-theory-driven metrics of training load are being commercially presented, requiring researchers and practitioners to investigate their relevance within any causal frameworks of interest. There is often no explicitly stated theoretical framework underlying the equations used or linking them to the mechanisms they are proposed to reflect.

The metrics also differ in their use, or not, of a particular threshold. For example, Dynamic Stress Load includes all peak resultant accelerations ≥ 2 *g*, whereas Body Load includes all time points where resultant acceleration ≥ 1.25 *g*. This may be an attempt to avoid the inclusion of random noise or very low impacts within the cumulative calculations. The specificity of these thresholds to the between-scapulae measurement site should be considered when interpreting studies that have applied the same processes at alternative body sites. For example, single-leg drop landings from a height of 0.45 m have resulted in peak surface-measured accelerations on average of 68 *g* at the metatarsophalangeal joint but 4 *g* at the C6 vertebra (129). Peak accelerations at the C6 vertebra level of 2.6 ± 0.8 *g* following 0.15-m single-leg drop landings may not have been sufficient to exceed the abovementioned thresholds in some individuals. This is an area that speaks to the requirement for further research to better understand what is measured and included in the approximation or accumulation of accelerations expressed in these accelerometer load metrics. Practitioners would be better able to make informed decisions with a better understanding of the quantification of these arbitrary unit metrics.

Validity and Reliability

The manufacturers and metrics will be considered individually because of differences in the algorithms, unknown variations in signal processing before metric calculation, or differences caused by the application of the same algorithm to different manufacturers' accelerometers (139). The validity is therefore context specific, in that any metric can be assessed for a specific purpose but not extrapolated to all possible applications of that metric (64,170). In addition, data validity will depend on the technical specifications of the sensor, the accuracy of sensor calibration, sensor placement and fixation, and signal processing methods, among other factors (95).

Within a causal framework, validation requires that the metric is a valid reflection of the proposed causal mechanism linked to the outcome (i.e., construct validity) and that the proposed causal mechanism is indeed causally related to the outcome of interest (100). Practitioners should additionally be aware of the reliability of the measurement system consisting of device, processing, and calculation (169). Differences between individuals, sessions, or drills can only meaningfully be interpreted within the context of intraunit or interunit and intraindividual or /interindividual reliability, which should help inform best practice for data collection and processing methods in both research and applied settings (1,10). As a result, it is important to review the validity of raw acceleration measures; the reliability of raw acceleration measures; the reliability of any metrics calculated from the measured accelerations; how these metrics relate to other parts of any potential performance or injury-related causal framework; what such metrics can or cannot do as part of any such applied framework; and what additional research is necessary to inform the design and application of these frameworks.

The intraclass correlation coefficient (ICC) assesses the agreement between data (e.g., interunit, interday, relationship to a gold standard measure) (112). The coefficient of variation (CV) is the ratio of the standard deviation to the mean (172). Throughout this review, ICC and other magnitudes of correlation will be interpreted as $0.0 \leq negligible < 0.3 \leq low < 0.5 \leq moderate < 0.7 \leq high < 0.9 \leq very\ high \leq 1.0$ (89,135). Coefficient of variation will be interpreted as $good \leq 5\% < moderate \leq 10\% < poor$ (168). However, the importance of context-specific evaluation such as absolute differences and their practical significance for the intended application within a causal framework should not be underestimated (87).

Validity of Upper Back–Mounted Global Navigation Satellite System–Housed Accelerometer Data

Any upper back–mounted accelerometer should be evaluated against either true acceleration of the unit or an estimated acceleration of the underlying structure it is intended to represent as part of the causal framework (e.g., trunk segment or whole-body CoM). Typically, upper back–mounted accelerometers have been

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Table 1

Manufacturer-designed metrics derived from the triaxial accelerometers embedded in upper back–mounted global navigation satellite system player tracking technology. (*g*, acceleration due to gravity, 9.81 m·s^{-2}).

Metric (manufacturer)	Equation	Details of calculation
PlayerLoad (Catapult Sports)	$PL = \sum \frac{\sqrt{(ax_n - ax_{n-1})^2 + (ay_n - ay_{n-1})^2 + (az_n - az_{n-1})^2}}{100}$ where ay,ax,az = acceleration in the x,y,z axes, and n = the time point (71,138)	Described as the square root of the sum of changes between adjacent time points in acceleration in all 3 individual axes, divided by a scaling factor of 100 to give a summation in arbitrary units (30,86,138,192)
Body Load (GPSports)	$Normalized\ Acceleration\ Vector\ (NAV) = \sqrt{a_x{}^2 + a_y{}^2 + a_z{}^2} - g$ $Body\ Load = \sum \frac{(NAV + NAV^3) \div frequency}{Exercise\ Factor}$ where a_x, a_y, a_z = acceleration in the x,y,z axes	Some early studies report using the aforementioned PlayerLoad equation (2,87). In others, a multiplication by subject body mass and a loading factor was used instead of the division using a denominator of 100 (52,122). The loading factor ensured a nonlinear relationship between trunk-measured accelerations and Body Load, as this reflected the magnitude of resultant accelerations, e.g., 5.0 $g \leq$ 6.0 g [*light impact*] = 1; 6.0 $g \leq$ 6.5 g = 2; 6.5 $\leq$ 7.0 g [*moderate impact*] = 3; 7.0 $\leq$ 8.0 g [*heavy impact*] = 4; 8.0 $\leq$ 10.0 g = 5; >10.0 g [*severe impact*] = 6 The manufacturers have since proposed "New Body Load," which subtracts g from the resultant acceleration before calculation, includes a linear and nonlinear component, and is scaled by both the accelerometer sampling frequency and an unknown manufacturer-determined "exercise factor" (83,146,191). This is summed for all time points with a resultant acceleration ≥1.25 g and reported in arbitrary units
Dynamic Stress Load (STATSports)	$DSL = \sum_{i=1}^{n} peak\ acc_i^k \times SF$ where *peak acc* is the maximum resultant acceleration ≥2 g in a 0.1 s period. k is a weighting factor and SF is a scaling factor—both are proprietary factors of the manufacturer (20)	Dynamic Stress Load calculates the accumulation of acceleration as a measure of athlete load in arbitrary units, calculated from "the weighted sum of all impacts above or equal to 2 g" (20,22,193). These "impacts" are classified as "maximum accelerometer magnitude values above 2 g in a 0.1 s period" to isolate the peak acceleration of each ground contact above the stated intensity threshold. The values are weighted using a nonlinear function, based on an assumed nonlinear relationship between impact magnitude and the consequences of that impact within the body; for example, this assumes that a peak acceleration of 4 g is associated with more than twice the consequences of a 2 g peak acceleration. Instantaneous values are summed to provide Dynamic Stress Load (DSL) over defined time periods (78,83)

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Table 2

Studies reporting validity of global navigation satellite system–housed accelerometers using raw and filtered raw data.*†

Article	Device	Criterion	Activity	Filter cutoff	Error	Correlation
Edwards et al. (61)	GPSports SPI HPU (100 Hz)	Full-body marker set—3D motion capture system (300 Hz)	All	Manufacturer proprietary filtering	WA: CV = 12.68% (*poor*)	ICC = 0.22 (*negligible*)
					P: CV = 7.62% (*moderate*)	ICC = 0.61 (*moderate*)
			Slow running (3.3 m·s^{-1})		WA: CV = 6.19% (*moderate*)	ICC = 0.89 (*high*)
					P: CV = 6.06% (*moderate*)	ICC = 0.86 (*high*)
			Medium running (5.0 m·s^{-1})		WA: CV = 5.12% (*moderate*)	ICC = 0.94 (*very high*)
					P: CV = 6.10% (*moderate*)	ICC = 0.85 (*high*)
			Fast running (6.7 m·s^{-1})		WA: CV = 16.20% (*moderate*)	ICC = 0.72 (*high*)
					P: CV = 13.12% (*poor*)	ICC = 0.78 (*high*)
Kelly et al. (109)	GPSports SPI-ProX II (100 Hz)	ADXL345 triaxial sensor (100 Hz)	*Static* (flat surface with no more than 2° axis deviation)	Raw	%Diff = 27.5 to 30.5	
			Dynamic (fixed oscillation point of a vibration platform)	Raw	%Diff = 32.0 to 35.0	
Nedergaard et al. (137)	Catapult MinimaxX S4 (100 Hz)	Ground reaction force–derived centre of mass acceleration	Straight line run	Raw		*r* = 0.26 (*negligible*)
			45° CoD	Raw		*r* = 0.42 (*low*)
			90° CoD	Raw		*r* = 0.55 (*moderate*)
Roell et al. (162)	Catapult OptimEye S5 (100 Hz)	Three markers on device—3D motion capture system (200 Hz)	Sport-specific movements (series of drills involving acceleration and deceleration, 60°, 80°, 90°, and 360° CoD, and jumps)	*Complementary (accel. and gyroscope) filter*		
				100 Hz	Mean: CV = 7.34% (*moderate*)	*r* = 0.99 (*very high*)
					Peak: CV = 10.05% (*poor*)	*r* = 0.97 (*very high*)
				10 Hz	Mean: CV = 7.46% (*moderate*)	*r* = 0.98 (*very high*)
					Peak: CV = 10.63% (*poor*)	*r* = 0.97 (*very high*)
				5 Hz	Mean: CV = 10.78% (*poor*)	*r* = 0.96 (*very high*)
					Peak: CV = 15.98% (*poor*)	*r* = 0.95 (*very high*)
				Kalman Filter		
				100 Hz	Mean: CV = 5.99% (*moderate*)	*r* = 0.99 (*very high*)
					Peak: CV = 12.10% (*poor*)	*r* = 0.95 (*very high*)
				10 Hz	Mean: CV = 8.90% (*moderate*)	*r* = 0.98 (*very high*)
					Peak: CV = 12.74% (*poor*)	*r* = 0.94 (*very high*)
				5 Hz	Mean: CV = 16.48% (*poor*)	*r* = 0.90 (*very high*)
					Peak: CV = 22.56% (*poor*)	*r* = 0.85 (*high*)
Wundersitz et al. (204)	Catapult MinimaxX S4 (100 Hz)	Single marker on unit—3D motion capture system (200 Hz)	Walk (1.5 m·s^{-1})	10 Hz	CV = 6.5% (*moderate*)	
			Jog (3.3 m·s^{-1})	10 Hz	CV = 7.5% (*moderate*)	
			Run (5.0 m·s^{-1} female, 5.9 m·s^{-1} male)	10 Hz	CV = 9.3% (*moderate*)	
			Walk-jog-run	Raw	CV = 15.8% (*poor*)	
				20 Hz	CV = 13.8% (*poor*)	
				15 Hz	CV = 11.5% (*poor*)	
				10 Hz	CV = 8.9% (*moderate*)	
				8 Hz		

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Table 2
Studies reporting validity of global navigation satellite system–housed accelerometers using raw and filtered raw data.*† *(Continued)*

Article	Device	Criterion	Activity	Filter cutoff	Error	Correlation
					CV = 8.2% (*moderate*)	
				6 Hz	CV = 8.2% (*moderate*)	
Wundersitz et al. (205)	Catapult MinimaxX S4 (100 Hz)	Single marker on unit—3D motion capture system (500 Hz)	Physical collisions	Raw	CV = 15.1% (*poor*)	
				30 Hz	CV = 12.0% (*poor*)	
				25 Hz	CV = 10.8% (*poor*)	
				20 Hz	CV = 9.6% (*moderate*)	
				15 Hz	CV = 9.5% (*moderate*)	
				10 Hz	CV = 14.5% (*poor*)	
				8 Hz	CV = 19.3% (*poor*)	

*CoD = change of direction; WA = weight acceptance phase; P = propulsion phase; %Diff = percentage difference.
†Magnitude of correlations interpreted as 0.0 ≤ *negligible* < 0.30; 0.3 ≤ *low* < 0.50; 0.50 ≤ *moderate* < 0.70; 0.70 ≤ *high* < 0.90; *very high* ≥ 0.90 (93,139). Coefficient of variation (CV) interpreted as *good* ≤ 5% < *moderate* ≤ 10% < *poor* (173).

evaluated against 3D motion capture (61,204,205). Table 2 summarizes studies evaluating the validity of upper back–mounted accelerometers based on correlations between raw or filtered accelerometer data and 3D motion capture. Roell et al. (162) demonstrated that applying a complementary filter (combining accelerometer and gyroscope data) and resampling to 5 Hz decreased resultant acceleration root mean square error by 15% (0.52–0.44 $m \cdot s^{-2}$) for mean and 58% (3.30–1.40 $m \cdot s^{-2}$) for peak values in sport-specific movements. In summary, raw signal data from upper back–mounted GNSS units are generally less valid than filtered signal data for estimating trunk accelerations when assessed against 3D motion capture (Table 2) (89,135).

In addition, studies have assessed the capabilities of GNSS-embedded upper back–mounted accelerometers using correlations to GRF (Table 3), evaluating the relationship between accelerometer data and whole-body external biomechanical load (i.e., GRF parameters). It is noteworthy that the trunk represents the largest proportion of body mass (~50%), perhaps explaining why the segmental acceleration of the trunk often best represents the acceleration of whole-body CoM (137). Collectively, these studies demonstrate that although linear relationships between upper back–mounted accelerations and whole-body external biomechanical load may be present in team sports after appropriate processing (47), they are weak in some cases and so assumptions of simple linear relationships should be made with caution (137). If a measure of upper back–mounted acceleration is valid for its intended purpose, subsequent inferences depend on the reliability of those measurements. Any training load algorithms will ideally be applied to valid and reliable input data.

Overall, low-pass filtering at 8 Hz or 10 Hz for walking and jogging (CV: 6.5–7.5%, *moderate*) (204), 15 Hz or 20 Hz for physical collisions (CV: 9.5% *or* 9.6%, *moderate*) (205), and 20 Hz for continuous hopping, drop landing, and rebound jumps (r = 0.709 to 0.825, *high*) (176) have all been shown to improve the relationship between upper back–mounted accelerometers and 3D motion capture (204,205) or GRF (176). The general indication is that upper back–mounted accelerometers embedded in GNSS units can provide valid estimations of trunk or CoM accelerations when a complementary filter is applied or when the signal is filtered at 20 Hz for high-impact activities and 10 Hz for lower-impact activities. However, filtering decisions are again dependent on the purpose of data collection and desired relationships with other factors within an overall framework. If the purpose is not to track segmental kinematics but to monitor the postimpact elastic wave accelerations, filtering decisions should perhaps be made in relation to the frequency content of the signal component of interest (84,103,129,174).

Reliability of Global Navigation Satellite System–Housed Upper Back–Mounted Accelerometer Data

Comparatively, studies investigating the reliability of GNSS-housed upper back–mounted accelerometer raw data are relatively scarce, with most studies preferring to evaluate manufacturer accelerometer-based metrics. This could be attributed to the relative ease of obtaining and interpreting manufacturer-processed data as opposed to raw accelerometer data from the GNSS units. Nonetheless, Table 4 displays studies that have reported the reliability of data from GNSS-housed accelerometers. Overall, the reliability of GNSS-housed accelerometer data and metrics is largely *high* to *very high*, with predominantly *good* CV. However, some findings demonstrated only *moderate* correlations and *moderate* to *poor* CV. Therefore, practitioners should always consider the potential implications of the type of activity (e.g., static, dynamic, field-based) and surface (e.g., treadmill, track) on accelerometer data (27,79,80,109). If the context-specific validity and reliability of acceleration data are deemed acceptable, these can then be used to calculate training load metrics, which should themselves be assessed for reliability.

Table 5 displays studies reporting the reliability of accelerometer-based metrics, demonstrated by controlled shaker-table, standardized treadmill running, multidirectional soccer-specific tasks, and standardized soccer games. This review shows Catapult's PlayerLoad metric has been the most extensively assessed accelerometer-based metric for reliability. Overall findings suggest that PlayerLoad and some of its derived metrics can be reliable provided that the same device is used for repeated tests (15,16,139).

Convergent Validity of Accelerometer-Based Metrics

It can be difficult to assess the validity of PlayerLoad, Dynamic Stress Load, Body Load. or similar accelerometer-based metrics against internal biomechanical load during sporting activities, given that a true gold standard method would require invasive

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measures of continuous loading of internal structures. However, existing literature has attempted to validate some metrics using comparisons to variations of the same metric (to identify key features of the algorithm in a sensitivity analysis or determine the most effective version for a specific purpose) or assessed agreement relative to similar metrics. For example, Wylde et al. (206) compared PlayerLoad data from VXSport Log accelerometers with Catapult Optimeye S5 accelerometers (both 100 Hz), finding *high* to *very high* correlations between the devices for vertical, anterior-posterior, and mediolateral PlayerLoad (ICC: 0.970–0.956, 0.828; *high* to *very high*) (100).

Two related studies assessing tackling impacts in rugby league highlighted lower sensitivity to changes in mediolateral and anteroposterior axis accelerations because of high vertical accelerations (50,132). Although Hollville et al. (91) recognized the potential for limitations in the PlayerLoad algorithm given that it may be influenced by rotational changes of the reference frame potentially leading to erroneous calculations. These Body Load and PlayerLoad studies suggest that accelerometer-based metrics may be more valid for some specific purposes when the larger vertical acceleration component cannot mask smaller changes in other axes, such as horizontal collision accelerations masked by running foot strikes (50,55,132), or when the purely rotational components are disregarded (91). This may need to be considered in contexts where horizontal accelerations are of primary interest; however, it does need further research to be fully understood.

Further to these studies, PlayerLoad, Body Load, and Dynamic Stress Load have all been assessed for correlation with various biomechanical and physiological external and internal load measures. Consequently, this review considers the convergent validity of these metrics, referring to how closely one metric is associated with other measures of the same construct (e.g., other internal and external training load measures or parameters that one would expect to be associated with a valid measure of biomechanical training load). This can inform decisions regarding the validity of that metric as an operationalization of the concept of biomechanical training load, although interpreting the metrics as true mechanical constructs should be cautioned against.

Table 3

Studies reporting correlations between raw and filtered global navigation satellite system–housed accelerometer data and ground reaction force parameters using raw and filtered data.*†

Article	Device	Criterion	Activity	Filter cutoff	CV (%)	Correlation
Edwards et al. (61)	GPSports SPI HPU (100 Hz)	Vertical ground reaction force (1200 Hz)	All	Manufacturer-specific filter	WA: 12.68 (*poor*)	ICC = −0.11 (*negligible*)
					P: 7.62 (*moderate*)	ICC = −0.32 (*low*)
			Slow running (3.3 $m \cdot s^{-1}$)		WA: 6.19 (*moderate*)	ICC = 0.47 (*low*)
					P: 6.06 (*moderate*)	ICC = −0.36 (*low*)
			Medium running (5.0 $m \cdot s^{-1}$)		WA: 5.12 (*moderate*)	ICC = 0.50 (*moderate*)
					P: 6.10 (*moderate*)	ICC = −0.15 (*negligible*)
			Fast running (6.7 $m \cdot s^{-1}$)		WA: 16.20 (*poor*)	ICC = 0.31 (*low*)
					P: 13.12 (*poor*)	ICC = −0.02 (*negligible*)
Simons and Bradshaw (176)	Catapult MinimaxX S4 (100 Hz)	Vertical ground reaction force (500 Hz)	Continuous hopping	50 Hz		r = 0.818 (*high*)
				20 Hz		r = 0.825 (*high*)
				15 Hz		r = 0.825 (*high*)
				8 Hz		r = 0.860 (*high*)
			Drop landing	50 Hz		r = 0.699 (*moderate*)
				20 Hz		r = 0.727 (*high*)
				15 Hz		r = 0.720 (*high*)
				8 Hz		r = 0.746 (*high*)
			Rebound jump	50 Hz		r = 0.734 (*high*)
				20 Hz		r = 0.734 (*high*)
				15 Hz		r = 0.706 (*high*)
				8 Hz		r = 0.613 (*moderate*)
Wundersitz et al. (203)	GPSports SPI Pro (100 Hz)	Resultant ground reaction force (100 Hz)	Straight line run	Raw	16.4 (*poor*)	r = 0.31 (*low*)
				10 Hz	11.7 (*poor*)	r = 0.76 (*high*)
			45 CoD	Raw	16.6 (*poor*)	r = 0.00 (*negligible*)
				10 Hz	14.5 (*poor*)	r = 0.67 (*moderate*)
			90 CoD	Raw	18.1 (*poor*)	r = 0.48 (*low*)
				10 Hz	17.2 (*poor*)	r = 0.47 (*low*)
			180 CoD	Raw	23.0 (*poor*)	r = 0.30 (*low*)
				10 Hz	23.9 (*poor*)	r = 0.23 (*negligible*)

*CoD = change of direction; WA = weight acceptance phase; P = propulsion phase.

†Magnitude of correlations interpreted as 0.0 ≤ *negligible* < 0.30; 0.3 ≤ *low* < 0.50; 0.50 ≤ *moderate* < 0.70; 0.70 ≤ *high* < 0.90; *very high* ≥ 0.90 (93,139). Coefficient of variation (CV) interpreted as *good* ≤ 5% < *moderate* ≤ 10% < *poor* (173).

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Table 4

Studies reporting reliability of global navigation satellite system–housed accelerometer data.*

Article	Device	Activity	Acceleration measure	CV (%)	Correlation
Gomez-Carmona et al. (81)	RealTrack WIMU PRO (1000 Hz)	*Between-device*	Resultant	0.38 (*good*)	
		Static		0.78 (*good*)	
		Static with active intervals		0.59 (*good*)	
		Dynamic—hydraulic shaker 10 Hz		0.66 (*good*)	
		Dynamic—hydraulic shaker 30 Hz		2.05 (*good*)	
		Treadmill incremental running		2.46 (*good*)	
		SAFT90 soccer simulation drill			
		Within-Device	Resultant	0.23 (*good*)	
		Static		0.40 (*good*)	
		Static with active intervals		0.55 (*good*)	
		Dynamic—hydraulic shaker 10 Hz		0.12 (*good*)	
		Dynamic—hydraulic shaker 30 Hz			
Kelly et al. (109)	GPSports SPI-ProX II (100 Hz)	Mechanical device capable of reproducing highly consistent impacts	Raw peak acceleration	1.87–2.21 (*good*)	
Nicolella et al. (139)	Catapult OptimEye S5 (100 Hz)	*Anterior-posterior*	Peak		
		Shaker table oscillating at 3.0 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Shaker table oscillating at 1.0 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Shaker table oscillating at 0.5 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Shaker table oscillating at 0.1 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Mediolateral	Peak		
		Shaker table oscillating at 3.0 g			ICC = 0.77 (0.62–0.89) (*moderate* to *high*)
		Shaker table oscillating at 1.0 g			ICC = 0.96 (0.92–0.98) (*very high*)
		Shaker table oscillating at 0.5 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Shaker table oscillating at 0.1 g			ICC = 1.00 (0.99–1.00) (*very high*)
		Vertical	Peak		
		Shaker table oscillating at 3.0 g			ICC = 1.00 (1.00–1.00) (*very high*)
		Shaker table oscillating at 1.0 g			ICC = 1.00 (1.00–1.00) (*very high*)
		Shaker table oscillating at 0.5 g			ICC = 1.00 (1.00–1.00) (*very high*)
		Shaker table oscillating at 0.1 g			ICC = 1.00 (1.00–1.00) (*very high*)

*Magnitude of correlations interpreted as $0.0 \leq$ *negligible* < 0.30; $0.3 \leq$ *low* < 0.50; $0.50 \leq$ *moderate* < 0.70; $0.70 \leq$ *high* < 0.90; *very high* ≥ 0.90 (93,139). Coefficient of variation (CV) interpreted as *good* $\leq 5\% <$ *moderate* $\leq 10\% <$ *poor* (173).

Relationships With External Load Measures. Relationships between accelerometer-based metrics and exercise demand has been thoroughly researched. The correlations reported in this review for the metrics and algorithms demonstrate largely *moderate* to *very high* correlations with known external load measures suggesting the potential for acceleration accumulation metrics to be applied effectively for specific purposes within athlete load monitoring. There is evidence of Dynamic Stress Load being capable of distinguishing low, medium, and high training session intensities (67), increased load in competition compared with training (181), and a *small* difference between first team players and under-18 and under-23 age group players (157). Meanwhile, PlayerLoad has been shown to increase with treadmill running speed (16), the number and magnitude of rugby league collisions (71), and distance covered in soccer training (34). Together, this research suggests the potential for accelerometer-based metrics to distinguish between intensity levels in varied sports and activities, indicative of a certain level of convergent validity. However, researchers and practitioners should always consider their choice of measurement device or metric in relation to their specific objective.

Total Distance. Distance covered has been adopted as a performance and load monitoring parameter, used to assess playing standard (31,105), aerobic capacity (151,164), fatigue (6,133,211), and training loads (152,211). Multiple studies have assessed the relationship between PlayerLoad and distance covered, finding *high* to *very high* correlations ($r = 0.70$ to 0.98) in Australian football (11,17), rugby league (73,97), rugby union (160), soccer (34), and hockey (148). This is likely at least partly a result of an increased number of foot strikes when covering greater distances. Conversely, there is a very limited body of literature assessing the relationship between derivatives of Dynamic Stress Load and distance; 1 study reports no correlation ($r = -0.01$) with distance covered per minute in a study of 28 male futsal players (158). Current findings imply that factors other than running distance are likely determinants of accelerometer-based load metrics in sports such as futsal, consisting of frequent high-intensity accelerations, decelerations, and sport-specific actions that do not require running for long distances (13,116).

Running Accelerations and Decelerations. Accelerations and decelerations in running speed, beginning at 1 $m \cdot s^{-2}$ and exceeding 2 $m \cdot s^{-2}$ for at least half a second, are recognized contributors to overall mechanical load (85,86). Accelerations and decelerations have respectively contributed 7–10% and 5–7% of a total "player load" (a downscaled square of resultant acceleration) across all playing positions during 45 elite competitive soccer matches (54). It has been previously found that GPSports raw accelerometer data consistently overestimated average high-speed run acceleration from change in speed over 0–10 m and 10–20 m of a 20-m distance. This was attributed to the need to incorporate body mass and compensate for gravity to improve the accuracy and practicality of accelerometer-derived algorithms (5). However, Beato and Drust (20) found that maximal

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Table 5

Studies reporting reliability of accelerometer-based load metrics.*†

Article	Device	Activity	Metric	CV (%)	Correlation
Barreira et al. (15)	STATSports Viper (100 Hz)	Jogging	PL	14.5 (*poor*)	$r = 0.863$ (*high*)
			PL·min^{-1}	18.2 (*poor*)	$r = 0.903$ (*very high*)
		Side cut	PL	15.2 (*poor*)	$r = 0.892$ (*high*)
			PL·min^{-1}	17.8 (*poor*)	$r = 0.921$ (*very high*)
		Stride	PL	24.5 (*poor*)	$r = 0.831$ (*high*)
			PL·min^{-1}	21.2 (*poor*)	$r = 0.806$ (*high*)
		Sprint	PL	23.4 (*poor*)	$r = 0.949$ (*very high*)
			PL·min^{-1}	22.1 (*poor*)	$r = 0.865$ (*high*)
Barrett et al. (16)	Catapult MinimaxX S4 (100 Hz)	Two standardized incremental treadmill running tests (7–16 km·h^{-1})	PL	5.9 (*moderate*)	ICC = 0.93 (*very high*)
			PL anterior-posterior	9.1 (*moderate*)	ICC = 0.92 (*very high*)
			PL medio-lateral	12.0 (*poor*)	ICC = 0.80 (*high*)
			PL vertical	6.3 (*moderate*)	ICC = 0.93 (*very high*)
Beato et al. (23)	STATSports Apex (100 Hz)	Yo-Yo intermittent recovery level 1	Dynamic stress load		ICC = 0.681 (*moderate*)
Boyd et al. (27)	Catapult MinimaxX 2.0 (100 Hz)	*Between device*	PL		$r = 0.997$ (*very high*)
		Static		1.10 (*good*)	
		Hydraulic shaker at 0.5 g		1.04 (*good*)	
		Hydraulic shaker at 3.0 g		1.02 (*good*)	
		Australian football		1.94 (*good*)	
		Within device	PL		
		Static		1.01 (*good*)	
		Hydraulic shaker at 0.5 g		0.91 (*good*)	
		Hydraulic shaker at 3.0 g		1.05 (*good*)	
Clubb et al. (42)	Catapult OptimEye S5 (100 Hz)	7 v 7 + 6 game	PL		ICC = 0.79 (*high*)
			PL·min^{-1}		ICC = 0.79 (*high*)
			PL·m^{-1}		ICC = 0.79 (*high*)
			PL anterior-posterior		ICC = 0.39 (*low*)
			PL mediolateral		ICC = 0.59 (*moderate*)
			PL vertical		ICC = 0.56 (*moderate*)
		10 v 10 game	PL		ICC = 0.83 (*high*)
			PL·min^{-1}		ICC = 0.83 (*high*)
			PL·m^{-1}		ICC = 0.91 (*very high*)
			PL anterior-posterior		ICC = 0.66 (*moderate*)
			PL mediolateral		ICC = 0.79 (*high*)
			PL vertical		ICC = 0.80 (*high*)
		11 v 11 game	PL		ICC = 0.81 (*high*)
			PL·min^{-1}		ICC = 0.82 (*high*)
			PL·m^{-1}		ICC = 0.88 (*high*)
			PL anterior-posterior		ICC = 0.53 (*moderate*)
			PL mediolateral		ICC = 0.76 (*high*)
			PL vertical		ICC = 0.80 (*high*)
Gomez-Carmona et al. (80)	RealTrack WIMU PRO (1000 Hz)	Treadmill incremental running	PL	3.52 (*good*)	ICC = 0.93 (*very high*)
		Track IR	PL	6.57 (*moderate*)	ICC = 0.88 (*high*)
Nicolella et al. (139)	Catapult OptimEye S5 (100 Hz)	Shaker table oscillating at 3.0 g			
		Anterior-posterior	Catapult PL	Median 0.05 (*small*)	
			Calculated PL	Median 0.08 (*small*)	
		Mediolateral	Catapult PL	Median 0.06 (*small*)	
			Calculated PL	Median 0.36 (*small*)	
		Vertical	Catapult PL	Median 0.06 (*small*)	
			Calculated PL	Median 0.38 (*small*)	
Wylde et al. (206)	VX sport Log (100 Hz)	Badminton-specific incremental test	PL vertical	1.0 (*good*)	ICC = 0.996 (*very high*)
			PL anterior-posterior	2.3 (*good*)	ICC = 0.979 (*very high*)
			PL mediolateral	2.3 (*good*)	ICC = 0.958 (*very high*)

*IR = incremental running; PL = PlayerLoad.

†Magnitude of correlations interpreted as $0.0 \leq$ *negligible* < 0.30; $0.3 \leq$ *low* < 0.50; $0.50 \leq$ *moderate* < 0.70; $0.70 \leq$ *high* < 0.90; $0.90 \leq$ *very high* ≤ 1.0 (93,139). Coefficient of variation (CV) interpreted as *good* $\leq 5\% <$ *moderate* $\leq 10\% <$ *poor* (173).

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acceleration sprints elicited significantly greater Dynamic Stress Load ($d = 0.43$, *small*) than self-selected submaximal acceleration intensities within the same repeated sprint protocol (20). This is despite no significant differences in peak speed, average speed, or total distance between the 2 protocols, suggesting a potential association with external or internal forces and a capability to distinguish these intensities.

This is also supported by findings related to change of direction, which includes both CoM deceleration and acceleration, necessitating relatively high impulses and often associated with high peak GRF (173,199) and joint moments (25). Therefore, more intense or more frequent changes of direction would be expected to result in greater accelerometer-based metric values. Indeed, cumulative acceleration magnitude increases with the number of changes of direction within shuttle run protocols of equal total distance (182). Similarly, PlayerLoad increases with increased prescribed direction changes within a task matched for overall time and distance (97). The matching of overall time would have further augmented the acceleration and deceleration demands of the task beyond the isolated effects of the direction changes. Multiple studies have recognized that high-intensity actions over minimal distance, such as rapid accelerations and decelerations, and changes of direction, can have a significant contribution to an athlete's load (5,28,54,86). These actions may not be reflected in traditional distance-based GNSS metrics, such as distance covered (140,155) or running speed (94). Therefore, accelerometer metrics with the potential to quantify these loads could provide valuable insight into a significant contributor to overall athlete load.

Running Speed. The positive relationship between running acceleration and peak running speed suggests that accelerometer-based metrics may increase with running speed as with running acceleration (20,207). Indeed, overall and axis-specific PlayerLoad increased approximately linearly with treadmill running speed from 7 to 16 $km \cdot h^{-1}$ (16). PlayerLoad is also greater in repeated maximal sprints (mean peak speed: 6.9 ± 0.3 $m \cdot s^{-1}$) compared with intermittent incremental high-intensity running (6.2 ± 0.3 $m \cdot s^{-1}$) (78). These results were further supported by findings that peak instantaneous accelerations from GPSports units were greater during two 30-m sprints (8.28 ± 1.81 *g* and 8.33 ± 1.56 *g*) than two 10-m sprints (7.32 ± 2.25 *g* and 7.03 ± 1.82 *g*) (196). It is also noteworthy, however, that the detection of magnitudes above 5 *g* was more reliable during the 10-m sprint (CV: 4.7 *vs* 14.2%). The relationship between running speed and accelerometer-based metric values could be attributed to increased stride frequency and GRF in each axis at greater running speeds. However, differences in running kinematics mean that accelerometer-based metric results are not comparable between athletes and may vary with increasing fatigue (16).

Collisions and Impacts. Accurately quantifying collision loads is a common challenge for practitioners in contact sports. Collisions and impacts have been associated with significant increases in heart rate, session RPE, creatine kinase levels, decreases in upper-body neuromuscular function, decreased perception of well-being, increased fatigue, and possible injury risk (71,104,161). Although there is minimal evidence to evaluate the direct relationship between Dynamic Stress Load and collisions, Body Load has been positively evaluated (14) and applied (46,52) for the purpose of quantifying collisions in contact sports. Studies have also identified strong associations between PlayerLoad and collisions for type, magnitude, and frequency (71,73,160). However, some studies have used a derivative of PlayerLoad by isolating "PlayerLoadSlow" defined as the PlayerLoad at velocities <0.2 $m \cdot s^{-1}$. Roe et al. (160) reported *moderate* to *high* correlations with collision count when using PlayerLoad Slow for rugby union backs ($r = 0.613$ [*moderate*]) and forwards ($r = 0.701$ [high]), improving the correlation compared with standard PlayerLoad in backs ($r = 0.477$ [*low*]) and remaining high in forwards ($r = 0.727$ [*high*]).

Triaxial Body Load was found to be invalid for assessing tackling impacts in rugby league due to high vertical accelerations but was corrected by disregarding vertical data to generate a 2D Body Load metric. This suggested that large vertical accelerations may mask the horizontal accelerations from collisions (50,132). Because this may be attributed to high magnitudes of vertical force in a running foot strike, it stands to reason that PlayerLoadSlow may improve the quality of information gathered with vertical accelerations minimized (141,208). Conversely, Hulin et al. (97) reported a lower correlation for PlayerLoadSlow of $r = 0.47$ (*moderate*) in rugby league, suggesting that although it is promising, PlayerLoad and its derivatives should be used with caution when quantifying collision workloads, and consideration should be given to the activities being monitored.

Relationships With Internal Load Measures. There is a lack of research investigating the relationship between accelerometer-based load metrics and measures or estimates of internal biomechanical load, such as tissue-specific loads on the muscles, bones, tendons, and ligaments of the musculoskeletal system. This is mostly due to the difficulty in obtaining field-based estimates of these loading parameters (191). Recent methodological developments offer the potential for future research and developments in this area, which will be discussed later in this review. Therefore, this section focuses on studies investigating the relationship between accelerometer-based load metrics and measures of internal physiological load.

Overall, correlations with measures of internal physiological load have been mixed. *Low* and *negligible* correlations with blood lactate measures suggest that it is unlikely accelerometer-based load measures will coincide with biochemical blood markers (184), which is understandable given that lactate concentrations are unlikely to be primarily driven by mechanical load. However, there are *moderate* to *very high* correlations with heart rate measures (16,179) and TRIMP variations (149,167), and *moderate* to *high* correlations with high-intensity intermittent running capacity (final velocity within 30–15 intermittent fitness test) (149) and YoYo tests for aerobic capacity (113,142). These can likely be attributed to players with greater aerobic capacity being capable of maintaining greater workloads (166). There are some *negligible* to *moderate* correlations for Body Load and session RPE variations (83,119,197); however, there are *moderate* to *high* correlations with RPE and session RPE for PlayerLoad (34,66,75,131,167) and Dynamic Stress Load (60,183). This demonstrates a contrast between PlayerLoad and Dynamic Stress Load (*moderate* to *high*) and Body Load (*negligible* to *moderate*), possibly because of differences in either the equations applied or the activities investigated. However, regarding correlations with session RPE, it should be noted that even a completely uncorrelated $load \cdot min^{-1}$ and RPE may correlate significantly once both are multiplied by the same session duration (128). Although accelerometer loads cannot be perfectly correlated with internal physiological loads, there are examples of *moderate* to *very high* correlations across internal physiological load measures that may support the use of accelerometer-based

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load metrics as a tool for monitoring athlete load for specific purposes. However, practitioners should ensure that they are aware of the capabilities specific to the manufacturer and algorithms used, as well as alternative methods for measuring physiological loads directly or with greater construct validity.

Fatigue. Fatigue has been shown to influence running mechanics, resulting in increases in peak vertical impact forces and peak propulsive forces (19,41). Accelerometer-based metrics could provide valuable information if they can demonstrate or detect the onset or accumulation of fatigue. It has been suggested that PlayerLoad could reflect changes in mediolateral and anteroposterior plane movement quality resulting from fatigue (143). Training loads in a previous 2 and 7 day period, and so possible fatigue, have significantly affected Dynamic Stress Load in Gaelic football (48). In another study, absolute and relative (per minute) Dynamic Stress Load were *moderately* affected by fatigue (i.e., differences between the same protocol performed before and after maximal repeated sprints). Therefore, it appears possible that intraindividual differences in Dynamic Stress Load magnitudes could be used to indicate acute fatigue (23,48). However, there are relatively few studies investigating this relationship, and further research should be undertaken to establish causal relationships.

Injury. Although circumstance and mechanism of injury may differ, it is ultimately a result of applied forces, stresses, and strains that cause damage exceeding the strength and repairability of the tissue (12,62,107,108,130). As a result, mechanical loading is considered a key contributor to tissue damage accumulation and therefore athletic injuries (108). Knowledge of relationships between accelerometer-based metrics and injury risk would be particularly useful. A 4-week accumulation of PlayerLoad (but not 2- or 3-week accumulations) had a *moderate* association with injury probability in rugby league, estimated from injury occurrence {Area-Under-Curve (95% confidence interval [CI]) = 0.585 [0.482–0.672]; $p = 0.048$} (51). In addition, Ehrmann et al. (63) reported that players had significantly lower Body Load 1 week (15.4% lower) and 4 weeks (9.0%) before injury when compared with the season average. This was suggested to possibly reflect voluntary or involuntary reduced outputs before the injury due to fatigue or soreness. Thus, both increased and decreased accelerometer-based metric values have been associated with greater injury risk through retrospective analysis, highlighting the need for a clear causal framework and prospective investigations.

Furthermore, Wiig et al. (202) found PlayerLoad to be a poor predictor of muscle damage following competitive soccer. Despite this, the literature overall suggests some potential to apply accelerometer-based metrics toward injury monitoring. Research should establish further relationships between accelerometer-based cumulative load metrics and injury risk, such as in different environments, and considering other current accelerometer-based metrics, such as Dynamic Stress Load. It is likely that any such efforts will need to focus on specific injuries as part of a clear causal framework and quantify or predict tissue loading cycle magnitudes and frequencies as well as intercycle recovery periods (108).

Limitations of Global Navigation Satellite System–Housed Accelerometers for Load Monitoring. Overall, the body of evidence has largely shown upper back–mounted accelerometers to be valid and reliable, but this is dependent on multiple factors, such as the manufacturer, processing, filtering, and activity being monitored, as well as the intended application. The manufacturers and metrics discussed in this review must be considered individually because of variations in the processing of accelerometer signals before metric calculation and in the application of different manufacturer algorithms. Accelerometer-based metrics can provide valid estimations of specific types of load under specific circumstances but cannot directly assess internal or external forces or mechanical load with absolute accuracy (47,61,162,189). Typically, raw data overestimates instantaneous accelerations, with error increasing with acceleration magnitude (61,196,204). In addition, upper back–mounted GNSS units are always susceptible to some incidental harness movement (61,132,206). These issues can be partially accounted for by refined signal processing (*e.g.*, appropriate filtering), an area requiring further study. Interdevice reliability results mean that practitioners should make comparisons within device where possible and plan unit use accordingly (15,16,42,47,80,139). Therefore, practitioners should consider the implications of the data collection method, type of activity, and surface on accelerometer data. Testing and monitoring conditions should remain consistent where possible and avoid making between-athlete, manufacturer, or device comparisons. Finally, data processing pipelines should ideally include checks and countermeasures for sensor limitations or possible errors (31,95).

Caution is urged when interpreting accelerations from 1 or several segments as a measure of external whole-body biomechanical loading (191), as GNSS unit–derived accelerations neglect the influence of other body segment accelerations, considerable postimpact shock wave attenuation inferior to the sensor (117,129), and the contribution of various frequency components to the overall acceleration signal (129,174). Even an entirely accurate measure of GRF or CoM acceleration (i.e., external biomechanical training load) may not correlate with the internal forces experienced by specific tissues (i.e., internal biomechanical training load) if muscle forces are not accounted for within the causal framework (126). Combining intensity and duration to operationalize a single cumulative training load metric also has its limitations (56,100,145,156,175). The accumulation of training load over time fails to consider the pattern of that load, for example, constant or variable, number and magnitude of individual loading cycles, rest periods, and the like. Indeed, it may be better to consider intensity, duration, and the interaction between the 2 separately in an overall statistical model (56,100,175) or even in a time-varying manner that reflects the true dynamic nature of many biological systems (100,178). Finally, it should be noted that dose-response relationships between accelerometer-based metrics and other aspects of the biomechanical load-adaptation pathway (189) remain mostly unknown.

Suggestions for Future Research

For accelerometer-based metrics to effectively contribute to training prescription, injury risk reduction, or rehabilitation as part of a causal framework, it is necessary to quantify dose-response relationships with other aspects of the relevant biomechanical load-adaptation pathways. This includes knowledge of how to effectively manipulate accelerometer-based metrics through training design, as well as knowledge regarding the relationships between these external biomechanical training load metrics and subsequent internal biomechanical training load, tissue damage, and remodeling.

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One area of growing focus is the need and current relative inability to measure internal biomechanical training loads outside of a laboratory (191), against which to assess accelerometer-based metrics. This is particularly relevant when seeking to associate training load measures with injury likelihood or inform progressive overloading of the tissue during rehabilitation or injury risk reduction programs. Greater insight into internal biomechanical training load can perhaps be gained when accelerometer or inertial measurement unit data are used alongside existing models of tissue stress response or as inputs to neuromusculoskeletal models (4,108,195). Open-source methodologies exist for the use of multiple IMUs as input to whole-body musculoskeletal computer simulation models that could estimate internal biomechanical loads (4,177), but the number or size of sensors required will likely need to be reduced before any real-world application for training load monitoring. However, physics-based or machine learning models can be used to predict accurate lower limb bone loads from minimal wearable technology data (65,127). Alternatively, it may be possible to identify algorithms to recognize certain activities (38,77,124,146) and then scale tissue-specific load estimates from laboratory-based estimates using an intensity measure derived from GNSS units.

Any developed tools for the estimation of accurate external or internal biomechanical training loads may facilitate the prescription and monitoring of specific loads, as well as the ranking of exercises in terms of specific tissue loading (59,163), or the provision of haptic (57,76) or audio (29,187) biofeedback. Before evaluating new or existing metrics, it would be useful to quantify the desired level of validity or reliability (e.g., how accurate is accurate enough?). This will depend on the intended application. Finally, we should not only assess the metrics themselves but also the consequences of prescribing changes in those metrics.

Practical Applications

The available body of evidence demonstrates multiple potential effective applications for accelerometer-based load metrics in estimating and monitoring athlete load. However, researchers and practitioners using this technology should always consider their choice of measurement device or metric in relation to their specific objective. Practitioners applying these metrics should remain aware of the limitations, recognize what is known about the origin of these metrics, and ensure results are processed, applied, and interpreted with these in mind. For example, GNSS-housed accelerometer signals can be valid for estimating trunk accelerations when filtered appropriately; appropriate low-pass cutoff frequencies may be approximately 20 Hz for high-impact activities and 10 Hz for lower-impact activities. Upper back–mounted accelerometer signals can be affected by factors, such as incidental harness or attachment movement (61,132,206); therefore, sensor fixation should be as tight as is physically comfortable to minimize additional displacement (32,95,118,186). In addition, studies assessing reliability of accelerometer-based load metrics found that reliability can be reduced between devices, when compared between athletes, between activity types, and on different surfaces. As a result, it is recommended to maintain consistency in testing and monitoring conditions and allocate the same devices to the same athletes where possible. When interpreting accelerometer-based metrics, practitioners should regard the validity for purpose and relative error both within and between devices. Metric validity is dependent on the specific purpose, such as, in rugby collisions where the larger vertical acceleration component may mask accelerations in horizontal axes (50,55,132), the exclusion of vertical accelerations may improve the application. Although between-device reliability dictates that practitioners should minimize comparisons between different units used by the same athlete, avoid between-athlete comparisons because of potential differences in mass and technique, and avoid comparisons between manufacturers because of metric algorithm and processing variations (15,16,42,47,80,139). Practitioners can use the reported validity and reliability values to inform decisions regarding selection and application of metrics, and the determination of magnitudes of change necessary to be considered "meaningful" (i.e., greater than chance or random variation). An example of this can be seen in Swinton et al. (180), which demonstrates the conversion of CV into CI to identify "true score" ranges.

References

1. Agmon N, Ahituv N. Assessing data reliability in an information system. *J Manag Inf Syst* 4: 34–44, 1987.
2. Aguiar MVD, Botelho GMA, Gonçalves BSV, Sampaio JE. Physiological responses and activity profiles of football small-sided games. *J Strength Cond Res* 27: 1287–1294, 2013.
3. Akenhead R, Nassis GP. Training load and player monitoring in high-level football: Current practice and perceptions. *Int J Sports Physiol Perform* 11: 587–593, 2016.
4. Al Borno M, O'Day J, Ibarra V, et al. OpenSense: An open-source toolbox for inertial-measurement-unit-based measurement of lower extremity kinematics over long durations. *J NeuroEng Rehabil* 19: 22, 2022.
5. Alexander JP, Hopkinson TL, Wundersitz DWT, Serpell BG, Mara JK, Ball NB. Validity of a wearable accelerometer device to measure average acceleration values during high-speed running. *J Strength Cond Res* 30: 3007–3013, 2016.
6. Alghannam AF. Metabolic limitations of performance and fatigue in football. *Asian J Sports Med* 3: 65–73, 2012.
7. Andersen MB, Williams JM. A model of stress and athletic injury: Prediction and prevention. *J Sport Exerc Psychol* 10: 294–306, 1988.
8. Armitage M, McErlain-Naylor SA, Devereux G, et al. On-field rehabilitation in football: Current practice and perceptions. A survey of the English premier league and football league. *Sci Med Footb*: 1–10, 2024.
9. Aroganam G, Manivannan N, Harrison D. Review on wearable technology sensors used in consumer sport applications. *Sensors* 19: 1983, 2019.
10. Atkinson G, Nevill AM. Statistical methods for assessing measurement error (reliability) in variables relevant to sports medicine. *Sports Med* 26: 217–238, 1998.
11. Aughey RJ. Applications of GPS technologies to field sports. *Int J Sports Physiol Perform* 6: 295–310, 2011.
12. Bahr R, Krosshaug T. Understanding injury mechanisms: A key component of preventing injuries in sport. *Br J Sports Med* 39: 324–329, 2005.
13. Barbero-Alvarez JC, Soto VM, Barbero-Alvarez V, Granda-Vera J. Match analysis and heart rate of futsal players during competition. *J Sports Sci* 26: 63–73, 2008.
14. Barr M, Beaver T, Turczyn D, Cornish S. Validity and reliability of 15 Hz global positioning system units for assessing the activity profiles of university football players. *J Strength Cond Res* 33: 1371–1379, 2019.
15. Barreira P, Robinson MA, Drust B, Nedergaard N, Raja Azidin RMF, Vanrenterghem J. Mechanical Player Load using trunk-mounted accelerometry in football: Is it a reliable, task- and player-specific observation? *J Sports Sci* 35: 1674–1681, 2017.
16. Barrett S, Midgley A, Lovell R. PlayerLoad: Reliability, convergent validity, and influence of unit position during treadmill running. *Int J Sports Physiol Perform* 9: 945–952, 2014.
17. Bartlett JD, O'Connor F, Pitchford N, Torres-Ronda L, Robertson SJ. Relationships between internal and external training load in team-sport athletes: Evidence for an individualized approach. *Int J Sports Physiol Perform* 12: 230–234, 2017.

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18. Bataller-Cervero AV, Gutierrez H, DeRentería J, et al. Validity and reliability of a 10 Hz GPS for assessing variable and mean running speed. *J Hum Kinet* 67: 17–24, 2019.
19. Bazuelo-Ruiz B, Durá-Gil JV, Palomares N, Medina E, Llana-Belloch S. Effect of fatigue and gender on kinematics and ground reaction forces variables in recreational runners. *PeerJ* 6: e4489, 2018.
20. Beato M, Drust B. Acceleration intensity is an important contributor to the external and internal training load demands of repeated sprint exercises in soccer players. *Res Sports Med* 29: 67–76, 2021.
21. Beato M, Coratella G, Stiff A, Iacono AD. The validity and between-unit variability of GNSS units (STATSports apex 10 and 18 Hz) for measuring distance and peak speed in team sports. *Front Physiol* 9: 1288, 2018.
22. Beato M, Devereux G, Stiff A. Validity and reliability of global positioning system units (STATSports viper) for measuring distance and peak speed in sports. *J Strength Cond Res* 32: 2831–2837, 2018.
23. Beato M, De Keijzer KL, Carty B, Connor M. Monitoring fatigue during intermittent exercise with accelerometer-derived metrics. *Front Physiol* 10: 780, 2019.
24. Bergamini E, Melis M, Lentola A, Camomilla V. Estimate of trunk inclination during fast movements by inertial sensing. In: *31st International Conference on Biomechanics in Sports*. Shiang T-Y, Ho W-H, Chenfu Huang P, Tsai C-L, eds. Taipei, Taiwan: International Society of Biomechanics in Sports (ISBS), 2013.
25. Besier TF, Lloyd DG, Cochrane JL, Ackland TR. External loading of the knee joint during running and cutting maneuvers. *Med Sci Sports Exerc* 33: 1168–1175, 2001.
26. Bezodis NE, North JS, Razavet JL. Alterations to the orientation of the ground reaction force vector affect sprint acceleration performance in team sports athletes. *J Sports Sci* 35: 1817–1824, 2017.
27. Boyd LJ, Ball K, Aughey RJ. The reliability of MinimaxX accelerometers for measuring physical activity in Australian football. *Int J Sports Physiol Perform* 6: 311–321, 2011.
28. Boyd LJ, Ball K, Aughey RJ. Quantifying external load in Australian football matches and training using accelerometers. *Int J Sports Physiol Perform* 8: 44–51, 2013.
29. Brake Mte, Stolwijk N, Staal B, Van Hooren B. Using beat frequency in music to adjust running cadence in recreational runners: A randomized multiple baseline design. *Eur J Sport Sci* 23: 345–354, 2023.
30. Bredt SdGT, Chagas MH, Peixoto GH, Menzel HJ, de Andrade AGP. Understanding player load: Meanings and limitations. *J Hum Kinet* 71: 5–9, 2020.
31. Brewer C, Dawson B, Heasman J, Stewart G, Cormack S. Movement pattern comparisons in elite (AFL) and sub-elite (WAFL) Australian football games using GPS. *J Sci Med Sport* 13: 618–623, 2010.
32. Camomilla V, Dumas R, Cappozzo A. Human movement analysis: The soft tissue artefact issue. *J Biomech* 62: 1–4, 2017.
33. Camomilla V, Bergamini E, Fantozzi S, Vannozzi G. Trends supporting the in-field use of wearable inertial sensors for sport performance evaluation: A systematic review. *Sensors* 18: 873, 2018.
34. Casamichana D, Castellano J, Calleja-Gonzalez J, San Román J, Castagna C. Relationship between indicators of training load in soccer players. *J Strength Cond Res* 27: 369–374, 2013.
35. Castillo ER, Lieberman DE. Shock attenuation in the human lumbar spine during walking and running. *J Exp Biol* 221: jeb177949, 2018.
36. Castillo D, Weston M, McLaren SJ, Cámara J, Yanci J. Relationships between internal and external match-load indicators in soccer match officials. *Int J Sports Physiol Perform* 12: 922–927, 2017.
37. Caulfield B, Reginatto B, Slevin P. Not all sensors are created equal: A framework for evaluating human performance measurement technologies. *NPJ Digit Med* 2: 7, 2019.
38. Chambers R, Gabbett TJ, Cole MH, Beard A. The use of wearable microsensors to quantify sport-specific movements. *Sports Med* 45: 1065–1081, 2015.
39. Checkoway H, Pearce NE, Kriebel D. *Research Methods in Occupational Epidemiology*. New York, NY: Oxford University Press, 2004.
40. Chen KY, Bassett DR. The technology of accelerometry-based activity monitors: Current and future. *Med Sci Sports Exerc* 37: S490–S500, 2005.
41. Clansey AC, Hanlon M, Wallace ES, Lake MJ. Effects of fatigue on running mechanics associated with tibial stress fracture risk. *Med Sci Sports Exerc* 44: 1917–1923, 2012.
42. Clubb J, Towlson C, Barrett S. Measurement properties of external training load variables during standardised games in soccer: Implications for training and monitoring strategies. *PLoS One* 17: e0262274, 2022.
43. Colby MJ, Dawson B, Heasman J, Rogalski B, Gabbett TJ. Accelerometer and GPS-derived running loads and injury risk in elite Australian footballers. *J Strength Cond Res* 28: 2244–2252, 2014.
44. Connor M, Beato M, O'Neill M. Adaptive athlete training plan generation: An intelligent control systems approach. *J Sci Med Sport* 25: 351–355, 2022.
45. Cordier S, Stewart PA. Exposure assessment. *Handbook of Epidemiology*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2005. pp. 437–462.
46. Coughlan GF, Green BS, Pook PT, Toolan E, O'Connor SP. Physical game demands in elite rugby union: A global positioning system analysis and possible implications for rehabilitation. *J Orthop Sports Phys Ther* 41: 600–605, 2011.
47. Crang ZL, Duthie G, Cole MH, Weakley J, Hewitt A, Johnston RD. The validity and reliability of wearable microtechnology for intermittent team sports: A systematic review. *Sports Med* 51: 549–565, 2020.
48. Cullen BD, McCarren AL, Malone S. Ecological validity of self-reported wellness measures to assess pre-training and pre-competition preparedness within elite Gaelic football. *Sport Sci Health* 17: 163–172, 2021.
49. Cummins C, Orr R, O'Connor H, West C. Global positioning systems (GPS) and microtechnology sensors in team sports: A systematic review. *Sports Med* 43: 1025–1042, 2013.
50. Cummins C, McLean B, Conlan G, Coutts A. Influence of drill type and unit fitting on accelerometer derived loads in rugby league training. *J Sci Med Sport* 20: e117–e118, 2017.
51. Cummins C, Welch M, Inkster B, et al. Modelling the relationships between volume, intensity and injury-risk in professional rugby league players. *J Sci Med Sport* 22: 653–660, 2019.
52. Cunniffe B, Proctor W, Baker JS, Davies B. An evaluation of the physiological demands of elite rugby union using global positioning system tracking software. *J Strength Cond Res* 23: 1195–1203, 2009.
53. Cust EE, Sweeting AJ, Ball K, Robertson S. Classification of Australian football kick types in-situation via ankle-mounted inertial measurement units. *J Sports Sci* 39: 1330–1338, 2020.
54. Dalen T, Ingebrigtsen J, Ettema G, Hjelde GH, Wisløff U. Player load, acceleration, and deceleration during forty-five competitive matches of elite soccer. *J Strength Cond Res* 30: 351–359, 2016.
55. Davies MJ, Young W, Farrow D, Bahnert A. Comparison of agility demands of small-sided games in elite Australian football. *Int J Sports Physiol Perform* 8: 139–147, 2013.
56. de Vocht F, Burstyn I, Sanguanchaiyakrit N. Rethinking cumulative exposure in epidemiology, again. *J Expo Sci Environ Epidemiol* 25: 467–473, 2015.
57. DeJong Lempke AF, Stephens SL, Fish PN, et al. Sensor-based gait training to reduce contact time for runners with exercise-related lower leg pain: A randomised controlled trial. *BMJ Open Sport Exerc Med* 8: e001293, 2022.
58. Dellaserra CL, Gao Y, Ransdell L. Use of integrated technology in team sports: A review of opportunities, challenges, and future directions for athletes. *J Strength Cond Res* 28: 556–573, 2014.
59. Demangeot Y, Whiteley R, Gremeaux V, Degache F. The load borne by the achilles tendon during exercise: A systematic review of normative values. *Scand J Med Sci Sports* 33: 110–126, 2023.
60. Dowling J. *Long-term Associations between Subjective Ratings of Wellness and Exertion and Parameters of Training Load in Elite Soccer over Two Seasons*. Liverpool, UK: Liverpool John Moores University, 2019.
61. Edwards S, White S, Humphreys S, Robergs R, O'Dwyer N. Caution using data from triaxial accelerometers housed in player tracking units during running. *J Sports Sci* 37: 810–818, 2019.
62. Edwards WB. Modeling overuse injuries in sport as a mechanical fatigue phenomenon. *Exerc Sport Sci Rev* 46: 224–231, 2018.
63. Ehrmann FE, Duncan CS, Sindhusake D, Franzsen WN, Greene DA. GPS and injury prevention in professional soccer. *J Strength Cond Res* 30: 360–367, 2016.
64. Eignor DR. The standards for educational and psychological testing. In: *APA Handbook of Testing and Assessment in Psychology, Vol. 1: Test Theory and Testing and Assessment in Industrial and Organizational Psychology*. Washington, WA: American Psychological Association, 2013. pp. 245–250.
65. Elstub L, Nurse C, Grohowski L, Volgyesi P, Wolf D, Zelik KE. Tibial bone forces can be monitored using shoe-worn wearable sensors during running. *J Sports Sci* 40: 1741–1749, 2022.
66. Enes A, Oneda G, Alves DL, et al. Determinant factors of the match-based internal load in elite soccer players. *Res Q Exerc Sport* 92: 63–70, 2021.

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67. Esteves L, Santos P, Lago-Penas C. Relationship between training load indicators and training periodization during preseason in elite football goalkeepers. *Hum Mov* 2018: 89–97, 2018.
68. Evans SA, James DA, Rowlands D, Lee JB. Evaluation of accelerometer-derived data in the context of cycling cadence and saddle height changes in triathlon. *Sensors* 21: 871, 2021.
69. Fan Y. Dynamic principle of center of mass in human walking. *Br Biotechnol J* 3: 524–544, 2013.
70. Fortune E, Morrow MMB, Kaufman KR. Assessment of gait kinetics using triaxial accelerometers. *J Appl Biomech* 30: 668–674, 2014.
71. Gabbett T, Jenkins D, Abernethy B. Physical collisions and injury during professional rugby league skills training. *J Sci Med Sport* 13: 578–583, 2010.
72. Gabbett T, Sancho I, Dingenen B, Willy RW. When progressing training loads, what are the considerations for healthy and injured athletes? *Br J Sports Med* 55: 947–948, 2021.
73. Gabbett TJ. Relationship between accelerometer load, collisions, and repeated high-intensity effort activity in rugby league players. *J Strength Cond Res* 29: 3424–3431, 2015.
74. Gabbett TJ. Debunking the myths about training load, injury and performance: Empirical evidence, hot topics and recommendations for practitioners. *Br J Sports Med* 54: 58–66, 2020.
75. Gallo T, Cormack S, Gabbett T, Williams M, Lorenzen C. Characteristics impacting on session rating of perceived exertion training load in Australian footballers. *J Sports Sci* 33: 467–475, 2015.
76. Galofaro E, D'Antonio E, Lotti N, Masia L. Rendering immersive haptic force feedback via neuromuscular electrical stimulation. *Sensors* 22: 5069, 2022.
77. Ghosh I, Ramasamy Ramamurthy S, Chakma A, Roy N. DeCoach: Deep learning-based coaching for badminton player assessment. *Pervasive Mob Comput* 83: 101608, 2022.
78. Gibson NV, Henning G, Twist C. Movement characteristics, physiological and perceptual responses of elite standard youth football players to different high intensity running drills. *Sci Med Footb* 2: 281–287, 2018.
79. Gómez-Carmona CD, Bastida-Castillo A, García-Rubio J, Ibáñez SJ, Pino-Ortega J. Static and dynamic reliability of WIMU PRO accelerometers according to anatomical placement. *Proc Inst Mech Eng P J Sports Eng Technol* 233: 238–248, 2019.
80. Gómez-Carmona CD, Bastida-Castillo A, González-Custodio A, Olcina G, Pino-Ortega J. Using an inertial device (WIMU PRO) to quantify neuromuscular load in running: Reliability, convergent validity, and influence of type of surface and device location. *J Strength Cond Res* 34: 365–373, 2020.
81. Gómez-Carmona CD, Bastida-Castillo A, Ibáñez SJ, Pino-Ortega J. Accelerometry as a method for external workload monitoring in invasion team sports. A systematic review. *PLoS One* 15: e0236643, 2020.
82. Gómez-Carmona CD, Pino-Ortega J, Rico-González M. Microelectromechanical systems. In: *The Use of Applied Technology in Team Sport*. Pino-Ortega J, Rico-González M, eds. New York, NY: Routledge, 2021. pp. 50–71.
83. Gomez-Piriz PT, Jiménez-Reyes P, Ruiz-Ruiz C. Relation between total body load and session rating of perceived exertion in professional soccer players. *J Strength Cond Res* 25: 2100–2103, 2011.
84. Hamill J, Derrick TR, Holt KG. Shock attenuation and stride frequency during running. *Hum Mov Sci* 14: 45–60, 1995.
85. Harper DJ, Kiely J. Damaging nature of decelerations: Do we adequately prepare players? *BMJ Open Sport Exerc Med* 4: e000379, 2018.
86. Harper DJ, Carling C, Kiely J. High-intensity acceleration and deceleration demands in elite team sports competitive match play: A systematic review and meta-analysis of observational studies. *Sports Med* 49: 1923–1947, 2019.
87. Harrison AJ, McErlain-Naylor SA, Bradshaw EJ, et al. Recommendations for statistical analysis involving null hypothesis significance testing. *Sports Biomech* 19: 561–568, 2020.
88. Hausler J, Halaki M, Orr R. Application of global positioning system and microsensor technology in competitive rugby league match-play: A systematic review and meta-analysis. *Sports Med* 46: 559–588, 2016.
89. Hinkle DE, Wiersma W, Jurs SG. *Applied Statistics for the Behavioral Sciences* (5th ed.). Boston, MA: Houghton Mifflin, 2003.
90. Ho C-S, Chang C-H, Hsu Y-J, et al. Feasibility of the energy expenditure prediction for athletes and non-athletes from ankle-mounted accelerometer and heart rate monitor. *Sci Rep* 10: 8816, 2020.
91. Hollville E, Couturier A, Guilhem G, Rabita G. A novel accelerometry-based metric to improve estimation of whole-body mechanical load. *Sensors* 21: 3398, 2021.
92. Horsley BJ, Tofari PJ, Halson SL, et al. Does site matter? Impact of inertial measurement unit placement on the validity and reliability of stride variables during running: A systematic review and meta-analysis. *Sports Med* 51: 1449–1489, 2021.
93. Howe ST, Aughey RJ, Hopkins WG, Stewart AM, Cavanagh BP. Quantifying important differences in athlete movement during collision-based team sports: Accelerometers outperform Global Positioning Systems. *2017 IEEE International Symposium on Inertial Sensors and Systems (INERTIAL)*. Kauai, HI: IEEE, 2017. pp. 1–4.
94. Huggins RA, Giersch GEW, Belval LN, et al. The validity and reliability of global positioning system units for measuring distance and velocity during linear and team sport simulated movements. *J Strength Cond Res* 34: 3070–3077, 2020.
95. Hughes GTG, Camomilla V, Vanwanseele B, Harrison AJ, Fong DTP, Bradshaw EJ. Novel technology in sports biomechanics: Some words of caution. *Sports Biomech* 23: 393–401, 2024.
96. Hulin BT, Gabbett TJ, Kearney S, Corvo A. Physical demands of match play in successful and less-successful elite rugby league teams. *Int J Sports Physiol Perform* 10: 703–710, 2015.
97. Hulin BT, Gabbett TJ, Johnston RD, Jenkins DG. PlayerLoad variables: Sensitive to changes in direction and not related to collision workloads in rugby league match play. *Int J Sports Physiol Perform* 13: 1136–1142, 2018.
98. Impellizzeri FM, Rampinini E, Coutts AJ, Sassi A, Marcora SM. Use of RPE-based training load in soccer. *Med Sci Sports Exerc* 36: 1042–1047, 2004.
99. Impellizzeri FM, Rampinini E, Marcora SM. Physiological assessment of aerobic training in soccer. *J Sports Sci* 23: 583–592, 2005.
100. Impellizzeri FM, Shrier I, McLaren SJ, et al. Understanding training load as exposure and dose. *Sports Med* 53: 1667–1679, 2023.
101. Jackson BM, Polglaze T, Dawson B, King T, Peeling P. Comparing global positioning system and global navigation satellite system measures of team-sport movements. *Int J Sports Physiol Perform* 13: 1005–1010, 2018.
102. Jeffries AC, Marcora SM, Coutts AJ, Wallace L, McCall A, Impellizzeri FM. Development of a revised conceptual framework of physical training for use in research and practice. *Sports Med* 52: 709–724, 2022.
103. John D, Miller R, Kozey-Keadle S, Caldwell G, Freedson P. Biomechanical examination of the "plateau phenomenon" in ActiGraph vertical activity counts. *Physiol Meas* 33: 219–230, 2012.
104. Johnston RD, Gabbett TJ, Seibold AJ, Jenkins DG. Influence of physical contact on neuromuscular fatigue and markers of muscle damage following small-sided games. *J Sci Med Sport* 17: 535–540, 2014.
105. Johnston RJ, Watsford ML, Austin DJ, Pine MJ, Spurrs RW. Movement profiles, match events, and performance in Australian football. *J Strength Cond Res* 30: 2129–2137, 2016.
106. Johnston RD, Weaving D, Hulin BT, Till K, Jones B, Duthie G. Peak movement and collision demands of professional rugby league competition. *J Sports Sci* 37: 2144–2151, 2019.
107. Kalkhoven JT, Watsford ML, Impellizzeri FM. A conceptual model and detailed framework for stress-related, strain-related, and overuse athletic injury. *J Sci Med Sport* 23: 726–734, 2020.
108. Kalkhoven JT, Watsford ML, Coutts AJ, Edwards WB, Impellizzeri FM. Training load and injury: Causal pathways and future directions. *Sports Med* 51: 1137–1150, 2021.
109. Kelly SJ, Murphy AJ, Watsford ML, Austin D, Rennie M. Reliability and validity of sports accelerometers during static and dynamic testing. *Int J Sports Physiol Perform* 10: 106–111, 2015.
110. Kobsar D, Osis ST, Hettinga BA, Ferber R. Classification accuracy of a single tri-axial accelerometer for training background and experience level in runners. *J Biomech* 47: 2508–2511, 2014.
111. Kokshenev VB. Dynamics of human walking at steady speeds. *Phys Rev Lett* 93: 208101, 2004.
112. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med* 15: 155–163, 2016.
113. Lee M, Mukherjee S. Relationship of training load with high-intensity running in professional soccer players. *Int J Sports Med* 40: 336–343, 2019.
114. Lee TA, Pickard AS. Exposure definition and measurement. In: *Developing a Protocol for Observational Comparative Effectiveness Research: A User's Guide*. Velentgas P, Dreyer N, Nourjah P, eds. Rockville, MD: Agency for Healthcare Research and Quality (US), 2013.
115. Lee JB, Sutter KJ, Askew CD, Burkett BJ. Identifying symmetry in running gait using a single inertial sensor. *J Sci Med Sport* 13: 559–563, 2010.

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116. Leite WSS. Physiological demands in football, futsal and beach soccer: A brief review. *Eur J Phys Educ Sport Sci* 2: 1–9, 2016.
117. Liew BXW, Zhu X, Zhai X, McErlain-Naylor SA, McManus C. Association between fat and fat-free body mass indices on shock attenuation during running. *J Biomech* 165: 112025, 2024.
118. Liu T, Inoue Y, Shibata K. Measurement of soft tissue deformation to improve the accuracy of a body-mounted motion sensor. *J Med Dev Trans ASME* 3, 2009.
119. Lovell TWJ, Sirotic AC, Impellizzeri FM, Coutts AJ. Factors affecting perception of effort (session rating of perceived exertion) during rugby league training. *Int J Sports Physiol Perform* 8: 62–69, 2013.
120. Luczak T, Burch R, Lewis E, Chander H, Ball J. State-of-the-art review of athletic wearable technology: What 113 strength and conditioning coaches and athletic trainers from the USA said about technology in sports. *Int J Sports Sci Coach* 15: 26–40, 2020.
121. Lutz J, Memmert D, Raabe D, Dornberger R, Donath L. Wearables for integrative performance and tactic analyses: Opportunities, challenges, and future directions. *Int J Environ Res Public Health* 17: 59, 2019.
122. Macadam P, Cronin J, Neville J, Diewald S. Quantification of the validity and reliability of sprint performance metrics computed using inertial sensors: A systematic review. *Gait Posture* 73: 26–38, 2019.
123. Malone JJ, Lovell R, Varley MC, Coutts AJ. Unpacking the black box: Applications and considerations for using GPS devices in sport. *Int J Sports Physiol Perform* 12: S218–S226, 2017.
124. Mannini A, Sabatini AM. Machine learning methods for classifying human physical activity from on-body accelerometers. *Sensors* 10: 1154–1175, 2010.
125. Mannini A, Intille SS, Rosenberger M, Sabatini AM, Haskell W. Activity recognition using a single accelerometer placed at the wrist or ankle. *Med Sci Sports Exerc* 45: 2193–2203, 2013.
126. Matijevich ES, Branscombe LM, Scott LR, Zelik KE. Ground reaction force metrics are not strongly correlated with tibial bone load when running across speeds and slopes: Implications for science, sport and wearable tech. *PLoS One* 14: e0210000, 2019.
127. Matijevich ES, Scott LR, Volgyesi P, Derry KH, Zelik KE. Combining wearable sensor signals, machine learning and biomechanics to estimate tibial bone force and damage during running. *Hum Mov Sci* 74: 102690, 2020.
128. McErlain-Naylor SA, Beato M. Factors influencing the jump momentum—sprint momentum correlation: A data simulation. *Eur J Sport Sci* 22: 1847–1855, 2022.
129. McErlain-Naylor SA, King MA, Allen SJ. Surface acceleration transmission during drop landings in humans. *J Biomech* 118: 110269, 2021.
130. McIntosh AS. Risk compensation, motivation, injuries, and biomechanics in competitive sport. *Br J Sports Med* 39: 2–3, 2005.
131. McLaren SJ, Macpherson TW, Coutts AJ, Hurst C, Spears IR, Weston M. The relationships between internal and external measures of training load and intensity in team sports: A meta-analysis. *Sports Med* 48: 641–658, 2018.
132. McLean BD, Cummins C, Conlan G, Duthie G, Coutts AJ. The fit matters: Influence of accelerometer fitting and training drill demands on load measures in rugby league players. *Int J Sports Physiol Perform* 13: 1083–1089, 2018.
133. Milioni F, Vieira LHP, Barbieri RA, et al. Futsal match-related fatigue affects running performance and neuromuscular parameters but not finishing kick speed or accuracy. *Front Physiol* 7: 2016, 2016.
134. Moe-Nilssen R, Helbostad JL. Estimation of gait cycle characteristics by trunk accelerometry. *J Biomech* 37: 121–126, 2004.
135. Mukaka MM. Statistics corner: A guide to appropriate use of correlation coefficient in medical research. *Malawi Med J* 24: 69–71, 2012.
136. Nagahara R, Mizutani M, Matsuo A, Kanehisa H, Fukunaga T. Association of sprint performance with ground reaction forces during acceleration and maximal speed phases in a single sprint. *J Appl Biomech* 34: 104–110, 2018.
137. Nedergaard NJ, Robinson MA, Eusterwiemann E, Drust B, Lisboa PJ, Vanrenterghem J. The relationship between whole-body external loading and body-worn accelerometry during team-sport movements. *Int J Sports Physiol Perform* 12: 18–26, 2017.
138. Nedergaard NJ, Verheul J, Drust B, et al. The feasibility of predicting ground reaction forces during running from a trunk accelerometry driven mass-spring-damper model. *PeerJ* 6: e6105, 2018.
139. Nicolella DP, Torres-Ronda L, Saylor KJ, Schelling X. Validity and reliability of an accelerometer-based player tracking device. *PLoS One* 13: e0191823, 2018.
140. Nikolaidis PT, Clemente FM, van der Linden CMI, Rosemann T, Knechtle B. Validity and reliability of 10-hz global positioning system to assess in-line movement and change of direction. *Front Physiol* 9: 228, 2018.
141. Nilsson J, Thorstensson A. Ground reaction forces at different speeds of human walking and running. *Acta Physiol Scand* 136: 217–227, 1989.
142. Owen AL, Newton M, Shovlin A, Malone S. The use of small-sided games as an aerobic fitness assessment supplement within elite level professional soccer. *J Hum Kinet* 71: 243–253, 2020.
143. Page RM, Marrin K, Brogden CM, Greig M. Biomechanical and physiological response to a contemporary soccer match-play simulation. *J Strength Cond Res* 29: 2860–2866, 2015.
144. Paquette MR, Miller RH. Reconciling new with old injury paradigms and the need to dig deeper—comment on Nigg et al. *Curr Issues Sport Sci (CISS)*, 2018.
145. Passfield L, Murias JM, Sacchetti M, Nicolò A. Validity of the training-load concept. *Int J Sports Physiol Perform* 17: 507–514, 2022.
146. Perri T, Reid M, Murphy A, Howle K, Duffield R. Validating an algorithm from a trunk-mounted wearable sensor for detecting stroke events in tennis. *J Sports Sci* 40: 1168–1174, 2022.
147. Pino-Ortega J, Oliva-Lozano JM, Gantois P, Nakamura FY, Rico-González M. Comparison of the validity and reliability of local positioning systems against other tracking technologies in team sport: A systematic review. *Proc Inst Mech Eng P J Sports Eng Technol* 236: 73–82, 2022.
148. Polglaze T, Dawson B, Hiscock DJ, Peeling P. A comparative analysis of accelerometer and time–motion data in elite men's hockey training and competition. *Int J Sports Physiol Perform* 10: 446–451, 2015.
149. Rabbani A, Kargarfard M, Castagna C, Clemente FM, Twist C. Associations between selected training-stress measures and fitness changes in male soccer players. *Int J Sports Physiol Perform* 14: 1050–1057, 2019.
150. Rabita G, Dorel S, Slawinski J, et al. Sprint mechanics in world-class athletes: A new insight into the limits of human locomotion. *Scand J Med Sci Sports* 25: 583–594, 2015.
151. Radzimiński Ł, Szwarc A, Padrón-Cabo A, Jastrzębski Z. Correlations between body composition, aerobic capacity, speed and distance covered among professional soccer players during official matches. *J Sports Med Phys Fitness* 60: 257–262, 2020.
152. Rago V, Brito J, Figueiredo P, Krustrup P, Rebelo A. Relationship between external load and perceptual responses to training in professional football: Effects of quantification method. *Sports* 7: 68, 2019.
153. Rago V, Brito J, Figueiredo P, et al. Methods to collect and interpret external training load using microtechnology incorporating GPS in professional football: A systematic review. *Res Sports Med* 28: 437–458, 2020.
154. Randers MB, Nielsen JJ, Bangsbo J, Krustrup P. Physiological response and activity profile in recreational small-sided football: No effect of the number of players. *Scand J Med Sci Sports* 24: 130–137, 2014.
155. Rawstorn JC, Maddison R, Ali A, Foskett A, Gant N. Rapid directional change degrades GPS distance measurement validity during intermittent intensity running. *PLoS One* 9: e93693, 2014.
156. Renfree A, Casado A, McLaren S. Re-Thinking athlete training loads: Would you rather have one big rock or lots of little rocks dropped on your foot? *Res Sports Med* 30: 573–576, 2022.
157. Reynolds J, Connor M, Jamil M, Beato M. Quantifying and comparing the match demands of U18, U23, and 1ST team English professional soccer players. *Front Physiol* 12: 706451, 2021.
158. Ribeiro JN, Gonçalves B, Coutinho D, Brito J, Sampaio J, Travassos B. Activity profile and physical performance of match play in elite futsal players. *Front Psychol* 11: 1709, 2020.
159. Rispens SM, Cox LGE, Ejupi A, Delbaere K, Annegarn J, Bonomi AG. Validation of walking speed estimation from trunk mounted accelerometers for a range of walking speeds. *Sensors* 21: 1854, 2021.
160. Roe G, Halkier M, Beggs C, Till K, Jones B. The use of accelerometers to quantify collisions and running demands of rugby union match-play. *Int J Perform Anal Sport* 16: 590–601, 2016.
161. Roe G, Darrall-Jones J, Till K, et al. The effect of physical contact on changes in fatigue markers following rugby union field-based training. *Eur J Sport Sci* 17: 647–655, 2017.
162. Roell M, Mahler H, Lienhard J, Gehring D, Gollhofer A, Roecker K. Validation of wearable sensors during team sport-specific movements in indoor environments. *Sensors* 19: 3458, 2019.
163. Sancho I, Willy RW, Morrissey D, Malliaras P, Lascurain-Aguirrebeña I. Achilles tendon forces and pain during common rehabilitation exercises in male runners with Achilles tendinopathy. A laboratory study. *Phys Ther Sport* 60: 26–33, 2023.
164. Santos PJ, Valente AP, Soares JMC. Aerobic capacity versus total distance covered during a game in elite soccer players. *Med Sci Sports Exerc* 33: S157, 2001.

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165. Schelling X, Torres L. Accelerometer load profiles for basketball-specific drills in elite players. *J Sports Sci Med* 15: 585–591, 2016.
166. Schmitz B, Pfeifer C, Kreitz K, Borowski M, Faldum A, Brand S-M. The Yo-Yo intermittent tests: A systematic review and structured compendium of test results. *Front Physiol* 9: 870, 2018.
167. Scott BR, Lockie RG, Knight TJ, Clark AC, Janse de Jonge XAK. A comparison of methods to quantify the in-season training load of professional soccer players. *Int J Sports Physiol Perform* 8: 195–202, 2013.
168. Scott MTU, Scott TJ, Kelly VG. The validity and reliability of global positioning systems in team sport: A brief review. *J Strength Cond Res* 30: 1470–1490, 2016.
169. Scott PJ, Brown AW, Adedeji T, et al. A review of measurement practice in studies of clinical decision support systems 1998–2017. *J Am Med Inform Assoc* 26: 1120–1128, 2019.
170. Sechrest L. Validity of measures is No simple matter. *Health Serv Res* 40: 1584–1604, 2005.
171. Serpiello FR, Hopkins WG, Barnes S, et al. Validity of an ultra-wideband local positioning system to measure locomotion in indoor sports. *J Sports Sci* 36: 1727–1733, 2018.
172. Shechtman O. The coefficient of variation as an index of measurement reliability. In: *Methods of Clinical Epidemiology*. Heidelberg, Germany: Springer Berlin, 2013. pp. 39–49.
173. Shimokochi Y, Ide D, Kokubu M, Nakaoji T. Relationships among performance of lateral cutting maneuver from lateral sliding and hip extension and abduction motions, ground reaction force, and body center of mass height. *J Strength Cond Res* 27: 1851–1860, 2013.
174. Shorten MR, Winslow DS. Spectral analysis of impact shock during running. *Int J Sport Biomech* 8: 288–304, 1992.
175. Shrier I, Wang C, Stokes T, Trejovargas J, Stovitz SD, Steele RJ. Causal effects, workload and injury risk: The importance of specifying the research question. *J Sci Med Sport* 25: 574–578, 2022.
176. Simons C, Bradshaw EJ. Do accelerometers mounted on the back provide a good estimate of impact loads in jumping and landing tasks? *Sports Biomech* 15: 76–88, 2016.
177. Slade P, Habib A, Hicks JL, Delp SL. An open-source and wearable system for measuring 3D human motion in real-time. *IEEE Trans Biomed Eng* 69: 678–688, 2022.
178. Smith TJ, Kriebel D. Biologically based exposure assessment for epidemiology. In: *Current Topics in Occupational Epidemiology*. Croydon, Unite Kingdom: Oxford University Press, 2013. pp. 199–217.
179. Sparks M, Coetzee B, Gabbett TJ. Internal and external match loads of university-level soccer players: A comparison between methods. *J Strength Cond Res* 31: 1072–1077, 2017.
180. Swinton PA, Hemingway BS, Saunders B, Gualano B, Dolan E. A statistical framework to interpret individual response to intervention: Paving the way for personalized nutrition and exercise prescription. *Front Nutr* 5: 41, 2018.
181. Tallent J, Higgins M, Parker N, et al. Quantification of bowling workload and changes in cognitive function in elite fast bowlers in training compared with Twenty20 Cricket. *J Sports Med Phys Fitness* 59: 35–41, 2019.
182. Tang R, Murtagh C, Warrington G, et al. Directional change mediates the physiological response to high-intensity shuttle running in professional soccer players. *Sports* 6: 39, 2018.
183. Tang R. *The Influence of the Frequency of Directional Change on the Physiological Responses to Intermittent Exercise in Elite Football Players*. Liverpool, UK: Liverpool John Moores University, 2015.
184. Taylor RJ, Sanders D, Myers T, Abt G, Taylor CA, Akubat I. The dose-response relationship between training load and aerobic fitness in academy rugby union players. *Int J Sports Physiol Perform* 13: 163–169, 2018.
185. Theodoropoulos JS, Bettle J, Kosy JD. The use of GPS and inertial devices for player monitoring in team sports: A review of current and future applications. *Orthop Rev* 12: 7863, 2020.
186. Valiant GA, McMahon TA, Frederick E. In: *A New Test to Evaluate the Cushioning Properties of Athletic Shoes*. Biomechanics X-B, Jonsson B, eds. Champaign, IL: Human Kinetics, 1987. pp. 937–941.
187. Van den Berghe P, Derie R, Bauwens P, et al. Reducing the peak tibial acceleration of running by music-based biofeedback: A quasi-randomized controlled trial. *Scand J Med Sci Sports* 32: 698–709, 2022.
188. van der Worp H, Vrielink JW, Bredeweg SW. Do runners who suffer injuries have higher vertical ground reaction forces than those who remain injury-free? A systematic review and meta-analysis. *Br J Sports Med* 50: 450–457, 2016.
189. Vanrenterghem J, Nedergaard NJ, Robinson MA, Drust B. Training load monitoring in team sports: A novel framework separating physiological and biomechanical load-adaptation pathways. *Sports Med* 47: 2135–2142, 2017.
190. Verheul J, Gregson W, Lisboa P, Vanrenterghem J, Robinson MA. Whole-body biomechanical load in running-based sports: The validity of estimating ground reaction forces from segmental accelerations. *J Sci Med Sport* 22: 716–722, 2019.
191. Verheul J, Nedergaard NJ, Vanrenterghem J, Robinson MA. Measuring biomechanical loads in team sports—from lab to field. *Sci Med Footb* 4: 246–252, 2020.
192. Vickery WM, Dascombe BJ, Baker JD, Higham DG, Spratford WA, Duffield R. Accuracy and reliability of GPS devices for measurement of sports-specific movement patterns related to cricket, tennis, and field-based team sports. *J Strength Cond Res* 28: 1697–1705, 2014.
193. Waegli A, Skaloud J. Optimization of two GPS/MEMS-IMU integration strategies with application to sports. *GPS Solut* 13: 315–326, 2009.
194. Wakeling JM, Nigg BM, Rozitis AI. Muscle activity damps the soft tissue resonance that occurs in response to pulsed and continuous vibrations. *J Appl Physiol* 93: 1093–1103, 2002.
195. Wakeling JM, Febrer-Nafría M, De Groote F. A review of the efforts to develop muscle and musculoskeletal models for biomechanics in the last 50 years. *J Biomech* 155: 111657, 2023.
196. Waldron M, Worsfold P, Twist C, Lamb K. Concurrent validity and test–retest reliability of a global positioning system (GPS) and timing gates to assess sprint performance variables. *J Sports Sci* 29: 1613–1619, 2011.
197. Weaving D, Marshall P, Earle K, Nevill A, Abt G. Combining internal- and external-training-load measures in professional rugby league. *Int J Sports Physiol Perform* 9: 905–912, 2014.
198. Weaving D, Whitehead S, Till K, Jones B. Validity of real-time data generated by a wearable microtechnology device. *J Strength Cond Res* 31: 2876–2879, 2017.
199. Welinski ML, Lee LN, McBroom B, Mufarreh B, Gidley AD. Ground reaction forces and temporal characteristics define cutting performance. *Int J Exerc Sci* 14: 211–221, 2021.
200. Wellman AD, Coad SC, Goulet GC, McLellan CP. Quantification of competitive game demands of NCAA division I college football players using global positioning systems. *J Strength Cond Res* 30: 11–19, 2016.
201. White E, Armstrong BK, Saracci R. *Principles of Exposure Measurement in Epidemiology*. Oxford, UK: Oxford University Press, 2008.
202. Wiig H, Raastad T, Luteberget LS, Ims I, Spencer M. External load variables affect recovery markers up to 72 h after semiprofessional football matches. *Front Physiol* 10: 689, 2019.
203. Wundersitz DWT, Netto KJ, Aisbett B, Gastin PB. Validity of an upper-body-mounted accelerometer to measure peak vertical and resultant force during running and change-of-direction tasks. *Sports Biomech* 12: 403–412, 2013.
204. Wundersitz DWT, Gastin PB, Richter C, Robertson SJ, Netto KJ. Validity of a trunk-mounted accelerometer to assess peak accelerations during walking, jogging and running. *Eur J Sport Sci* 15: 382–390, 2015.
205. Wundersitz DWT, Gastin PB, Robertson SJ, Netto KJ. Validity of a trunk-mounted accelerometer to measure physical collisions in contact sports. *Int J Sports Physiol Perform* 10: 681–686, 2015.
206. Wylde MJ, Lee MBC, Chee Yong L, Callaway AJ. Reliability and validity of GPS-embedded accelerometers forthe measurement of badminton specific player load. *J Trainology* 7: 34–37, 2018.
207. Yıldız S, Ateş O, Gelen E, et al. The relationship between start speed, acceleration and speed performances in soccer. *Universal J Educ Res* 6: 1697–1700, 2018.
208. Yu L, Mei Q, Xiang L, et al. Principal component analysis of the running ground reaction forces with different speeds. *Front Bioeng Biotechnol* 9: 629809, 2021.
209. Zadpoor AA, Nikooyan AA. The relationship between lower-extremity stress fractures and the ground reaction force: A systematic review. *Clin Biomech* 26: 23–28, 2011.
210. Zandbergen MA, Ter Wengel XJ, van Middelaar RP, Buurke JH, Veltink PH, Reenalda J. Peak tibial acceleration should not be used as indicator of tibial bone loading during running. *Sports Biomech*: 1–18, 2023.
211. Zurutuza U, Castellano J, Echeazarra I, Casamichana D. Absolute and relative training load and its relation to fatigue in football. *Front Psychol* 8: 878, 2017.

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